Chapter test

13 Reproduction in flowering plants

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

★ 1. The photographs below show four plant structures.

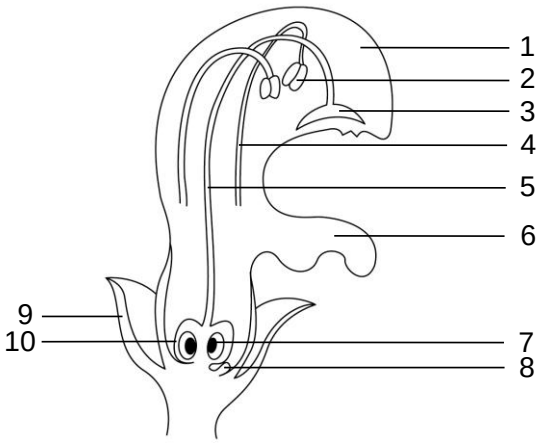
W ***X*** ***Y*** ***Z***



Which structure(s) is/are formed from sexual reproduction?

A. W only
B. X and Y only
C. W and Z only
D. W, Y and Z only ☐

Directions: Questions 2 to 4 refer to the diagram below, which shows the structures of a flower.



2. Meiosis occurs in

A. structures 2 and 10.
B. structures 2 and 3.
C. structures 2 and 7.
D. structures 3 and 7. ☐

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★ 3. Which of the following combinations about the pollination of this flower is correct?

	Agent of pollination	*Reason*
A.	insect	Structure 2 is enclosed by structure 1.
B.	insect	Structures 2 and 3 are present in the same flower.
C.	wind	Structure 8 is present.
D.	wind	Structure 3 is longer than structure 2.

☐

4. The zygote is formed in

A. structure 2.
B. structure 3.
C. structure 7.
D. structure 10. ☐

5. Which of the following structures in plants are involved in seed formation?

(1) sepals
(2) petals
(3) stamens
(4) carpels

A. (1) and (2)
B. (2) and (3)
C. (2) and (4)
D. (3) and (4) ☐

6. Which of the following combinations is ***incorrect***?

	Floral part	*Fate after fertilization*
A.	ovary wall	develops into the fruit wall
B.	ovule	develops into the fruit
C.	integument	develops into the seed coat
D.	zygote	develops into the embryo

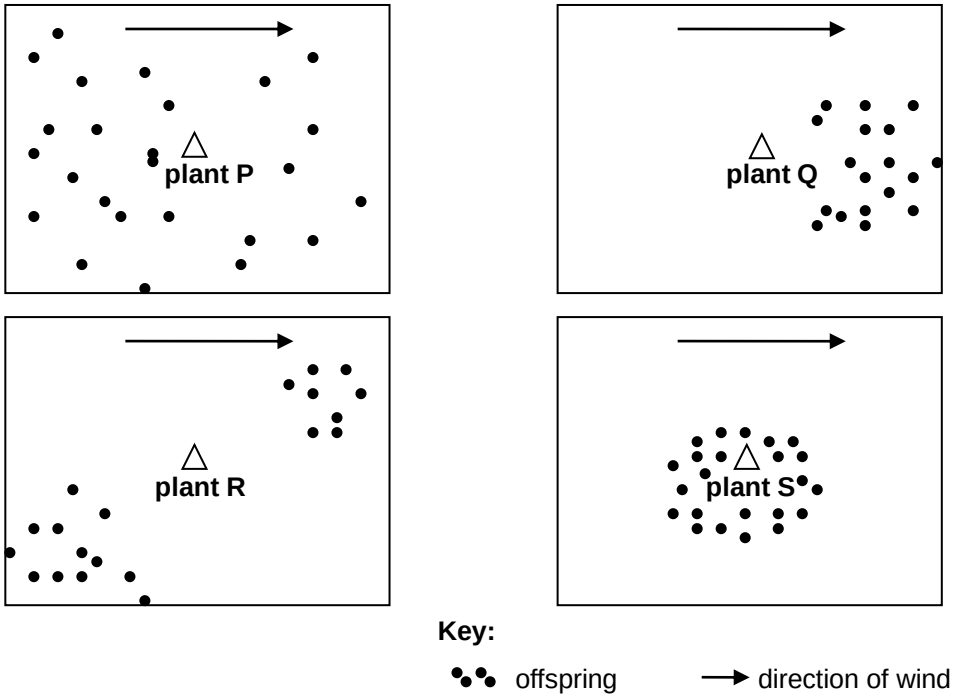
☐

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★ 7. Which of the following plant parts are genetically identical?

A. peas in the same pea pod
B. flesh and the seed coat of the same fruit
C. seed and the seed coat enclosing it
D. pollen grains produced by a flower ☐

8. The diagrams below show the positions of plants P, Q, R and S, and the distribution of their offspring respectively.



Which plant disperses its seeds by wind?

A. plant P B. plant Q
C. plant R D. plant S ☐

9. A new island was formed due to a volcanic activity. No species existed on the island. A seed was carried to the island by water currents. The seed grew and reproduced asexually. What are the advantages of carrying out asexual reproduction in this condition?

(1) The offspring can be produced quickly.
(2) The offspring are more able to adapt to environmental changes.
(3) Food reserves are available for the growth of the offspring.

A. (1) and (2) only B. (1) and (3) only
C. (2) and (3) only D. (1), (2) and (3) ☐

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★ 10. Potato plants can reproduce both sexually and asexually. Which of the following correctly compares the difference between these two types of reproduction?

		Sexual reproduction	***Asexual reproduction***
A.	Number of chromosomes in the cells of the offspring produced	haploid	diploid
B.	Level of competition between the offspring	higher	lower
C.	Region of colonization of the offspring	larger	smaller
D.	Genetic variation among the offspring	less	more

□

Question	1	2	3	4	5	6	7	8	9	10
Related section	13.2	13.3	13.3	13.3	13.3	13.3	13.3	13.3	13.4	13.4

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B. Problem solving (22 marks)

11. The photograph below shows the storage organ of a potato plant. Buds of the storage organ are growing.



(a) (i) What type of storage organ is shown in the photograph? (1 mark)

(ii) How can you test for the presence of the major food reserves stored in this storage organ? (2 marks)

(b) (i) Describe how potato plants form this type of storage organ. (2 marks)

(ii) Describe and explain the change in the size of the storage organ as the buds grow further. (2 marks)

(c) Is the production of new plants from storage organs a kind of asexual or sexual reproduction? Explain your answer. (2 marks)

(Total: 9 marks)

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★12. After fertilization, certain floral parts develop into seeds and fruits.

(a) State ***two*** advantages of dispersing the seeds and fruits far away from the parent plants. (2 marks)

(b) The photograph below shows two fruits.



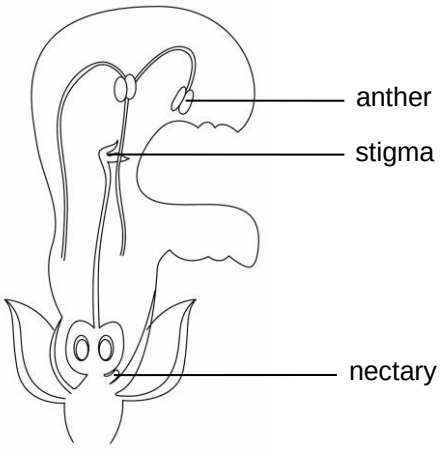
(i) Suggest the external agent required for the dispersal of these fruits. (1 mark)

(ii) With reference to the photograph, explain how these fruits are adapted for dispersing the seeds inside. (2 marks)

(Total: 5 marks)

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★ 13. The diagram below shows the section of a flower.



(a) Apart from the presence of nectaries, state ***two*** observable features from the diagram that suggest this flower is insect-pollinated. (2 marks)

(b) What is the significance of having different positioning of anthers and stigma in the flower? (2 marks)

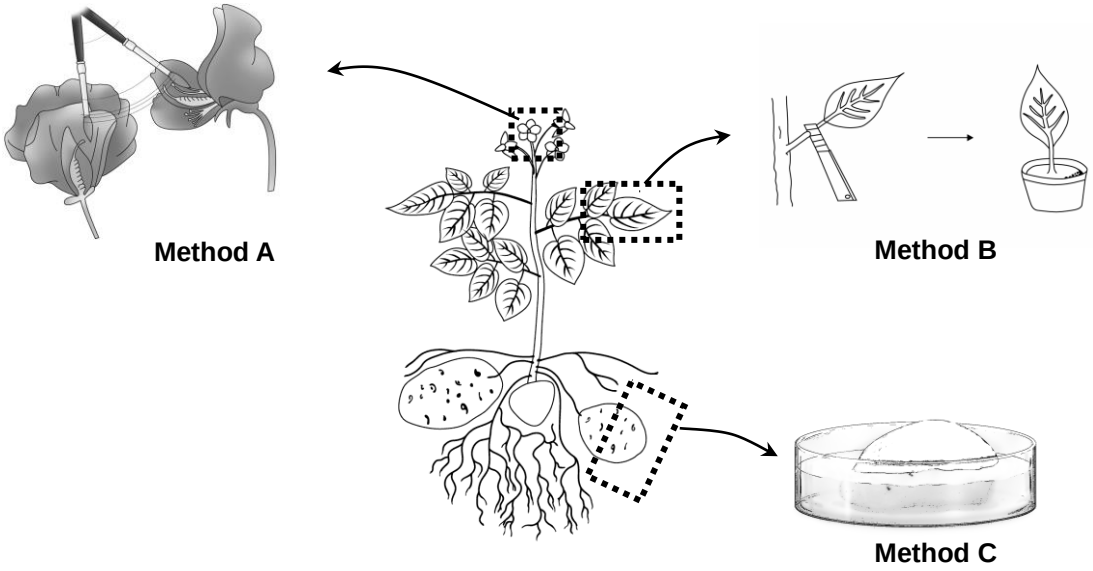
(c) Describe how fertilization takes place after pollination. (4 marks)

(Total: 8 marks)

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C. Exam challenge (8 marks)

★★14. A farmer would like to cultivate potatoes with high starch content. Potato plants could be cultivated artificially by the methods shown in the diagrams below. At the beginning, the farmer used Method A to propagate potato plants.



(a) (i) Give an advantage with reason of using Method A to propagate potato plants instead of using Methods B and C. (2 marks)

(ii) After several trials, the farmer found that potato plants produced by cross-pollination showed a higher degree of variety. Suggest the procedures he should do to ensure cross-pollination. (2 marks)

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(b) After cultivating many potato plants, the farmer finally produced a potato plant with very high starch content. Then he propagated this potato plant using Method C.

Explain why he used Method C to propagate this potato plant instead of using

(i) Method A. (2 marks)

(ii) Method B. (2 marks)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	C	3.	A	4.	C	5.	D
6.	B	7.	B	8.	B	9.	B	10.	C

(1 mark each)

B. Problem solving (22 marks)

11. (a) (i) Stem tuber (1)

(ii) Add iodine solution to a slice of potato tuber. (1)

If starch is present, the iodine solution turns from brown to blue-black. (1)

(b) (i) The green leaves of the potato plants carry out photosynthesis to produce food. (1)

Some food is passed underground to form tubers. (1)

(ii) The potato tuber becomes smaller. (1)

Some of the food stored in the tuber is used for the growth of the new plant. (1)

(c) It is a kind of asexual reproduction. (1)

This is because the process does not involve the fusion of gametes / fertilization. (1)

(Total: 9 marks)

12. (a) Any ***two*** of the following: (2)

- To avoid overcrowding (1)
- To reduce competition for resources (1)
- To colonize new areas (1)
- To prevent the spread of diseases or pests among plants (1)

(b) (i) Animals (1)

(ii) The fruits have hooks / spines, (1)

which allow them to stick to the fur of animals and are carried away by the animals. (1)

(Total: 5 marks)

13. (a) Large petals (1)

Anthers and stigma within the flower (1)

(b) To avoid self-pollination (1)

so that the genetic variation of the offspring can be increased. (1)

(c) After pollination, the pollen grain develops a pollen tube. (1)

The pollen tube secretes enzymes to digest the stigma tissue, creating a pathway through the style. (1)

The pollen tube carries the male gamete down the ovary. (1)

The male gamete is released into the ovule and fuses with the female gamete. (1)

(Total: 8 marks)

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C. Exam challenge (8 marks)

14. (a) (i) The daughter plants produced are genetically different from the parent plant. (1)
More variants can be produced to screen for potato plants which produce potatoes with high starch contents. (1)

(ii) Cut all the stamens of a potato flower. (1)
Transfer the pollen grains from a flower of another potato plant onto the stigma of the potato flower with its stamens cut. (1)

(b) (i) The offspring propagated by Method C is genetically identical to the parent while the offspring propagate by Method A is genetically different from the parent. (1)
The trait with high starch content can stably pass to the offspring by Method C. (1)

(ii) In Method C, food reserve is available for the growth of offspring, while there is no food reserve in Method B. (1)
Method C increases the chance of survival of the offspring. (1)

(Total: 8 marks)

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